

# Improving mechanical performance for extreme environments

*Prototype Project Report*

WSU Mechanics and Materials Lab



**WSUM-PythonApps**

Daniel Lee;

Daniel Book;

Douglas Takada;

[1.22.24]

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# *Introduction*

## **I. Project Introduction**

One of the many vital components to the space industry and defense is the protection against high velocity impact threats. It is crucial to know precisely how much damage manufactured protective shields can withstand before they break. Being able to accurately determine when a protective shield will fail will help these ultrahard ceramics and refractory alloys to be used safely and effectively to protect humans and equipment from impact threats.

Our team's aim is to develop a program that will analyze images of protective shields that have been damaged due to circumstances found in extreme environments, and then gather information about the shields. The main technique that we will be using is image segmentation. This will allow us to format the images in a way such that we will then be able to efficiently analyze them. By analyzing the damage as well as imperfections in the materials, we will be able to learn which circumstances allow for the best performance. This information will also be used in the development of stronger and more durable materials.

## **II. Background and Related Work**

The domain of this project can be defined as the “automated analysis of microscopic images from various materials in extreme environments.” Prior to automation, researchers relied on manual measurements and computations of their images to get the data they needed. This, however, proved to be increasingly difficult and costly over time, lacking a collection of big data to work with more effectively. Automation through image segmentation aims to streamline the data collection process and provide researchers with a means to analyze and interpret images in different ways.

One of the current image segmentation tools available for use is MATLAB (MathWorks, 2024), which provides users with applications that let them try out various image segmentation techniques such as thresholding and clustering. Their software also allows users to utilize deep learning called semantic segmentation to be able to train a network to apply what it has learned from a previous set of images onto new images. While MATLAB allows for the image segmentation tools to properly process images for their desired segments, it lacks a detailed feature for measurements such as size on those regions or statistical analysis on the data to determine various distributions. Our goal is to take a step beyond image segmentation and design for measurements and statistical analysis to provide a set of meaningful data for use in the domain.

Undertaking this project will require the learning of new skills and information fundamental to development as it progresses. At the core of this project is image segmentation, where it is used to identify and distinctly outline breakage patterns in a set of microscopic images. This will require an understanding of how image segmentation functions and the various segmentation methods available, along with the complex implementation in Python. Additionally, we need to perform statistical analysis on the segmented images to calculate various distribution measurements, where we will need to know how the data should be processed/formatted along with the various formulas and techniques. Finally, while our project does not directly extend our focus into the mechanical and physical aspect of materials engineering, a deeper understanding

and interest in the research can help us to design the project more effectively and better tailor to our client's needs.

### **III. Project Overview**

Testing materials with load until they break has been a common testing method throughout the research world. Our project will be focusing on materials in extreme environments such as intense temperatures or materials put under intense loads. Understanding the load that ultrahard ceramics and refractory alloys can sustain is crucial to improving the material to prevent or prolong its failure. This research will also help with real-world applications so that these materials are applied safely and efficiently. The WSUM-PythonApps team aims to develop a program package that will expedite the research process into these materials. Although in this project we will be focusing on boron carbide (B<sub>4</sub>C) ceramics and niobium-titanium (Nb-Ti) alloys, the grand vision for this application is for it to be easily applied to other materials.

For these materials, multiscale analysis is performed. Our scope of this project, however, will focus on the microstructures of the materials from electronic microscopic images. The combination of analysis at the microscopic level as well as at other scales, the research team can learn meaningful information about the material to alter its structural behavior.

In doing this, there are key objectives that are essential to this project. The first of these is to determine the best solution to identifying the pores of the material and calculating the most accurate measurements of each pore. Dr. Zare's research team has found promising results from using image segmentation to identify pores of the material and calculating correct measurements. This is done through analyzing the contrast of height of the rim and the inside of the pore of the electronic microscopic images. This contrast is what is used in the image segmentation and to calculate the measurements. However, we were encouraged to explore alternative solutions to find more accurate methods of calculating these measurements. The primary measurements that we will be calculating are the size, shape and spacing of each pore.

Once we can calculate accurate measurements of the pores, the second core objective is to apply a machine learning (ML) algorithm and statistical analysis to find meaningful information about the pores. With this information, the goal is to find relationships between mechanical properties and the pore structure. In this objective we are also hoping to look into various ML algorithms and statistical analysis techniques to find the most accurate model for our pictures. In addition, we will be analyzing the best types of electronic microscopic images that should be used to provide a more accurate analysis.

This project will be a continuation from Dr. Zare's student's previous work. We will be analyzing his program and developing our own programs to strive for high accuracy. For pore identification and measurements. We will primarily use Python. For the Statistical analysis and machine learning algorithms, we plan to use MATLAB and a python library called OpenCV.

Finally, our last key objective is to add everything into a package that makes calculating pore measurements and analyzing the structure simple and quick among mechanical and material engineers. For our implementation of this package, we are looking into a web application implementation, so it is easily accessible and plan to make a User Interface (UI) that is simple, easy to use, and intuitive for those who do not have a background in Computer Science.

The agreed upon timeline for these objectives are for the identification, measurement, and the statistical analysis of the pores to be completed by the end of the first semester; and the Graphical User Interface (GUI)/webpage to be implemented in the second half of the semester.

All the objectives we have stated should be shaped and molded to satisfy new objectives and requirements. As this is an application for a research lab, the approach that we take might vary depending on the client preference and an approach that yields a more accurate result.

#### **IV. Client and Stakeholder Identification and Preferences**

Our primary client is Dr. Zare of the Multiscale Mechanics and Materials Laboratory at Washington State University. This project will be used to assist Dr. Zare in the analysis of damage to niobium materials used in extreme environments, as well as the analysis of imperfections in those materials. As of right now there are not any other stakeholders in specific, but it is likely that there will be other collaborators in the near future who will use this project for similar analytics.

Furthermore, this project has the potential to expand to other areas of research. This would include other collaborators and scientists using the software to analyze a wider range of materials. This would also include expanding upon the image segmentation technologies used such that they can work efficiently in different, but similar use cases.

Our stakeholders will benefit from this project as it will result in a new and efficient way to analyze important data regarding the integrity of these materials in extreme environments. On top of that, the eventual implementation of a graphical user interface into this project will allow a much wider variety of users to use this technology without the need for a background in Python programming.

#### **V. Glossary**

Image segmentation - The process of partitioning a digital image into multiple image segments.

#### **VI. References**

[1]

"Image Segmentation," *MathWorks*. <https://www.mathworks.com/discovery/image-segmentation.html#image-segmentation-with-matlab> (accessed Jan. 25, 2024).